Equipment list

Flash drum D-400 unit · equipment, valves and relief devices with design conditions, materials and weights

Document no. FD-V1-ME-EQL-001

Every item of the D-400 unit with its type, size, operating and design conditions, material, corrosion allowance, insulation, thickness and weight, from the heat and material balance FD-V1-PE-HMB-001 and the design-basis rules. P-400 is listed as an interface item (upstream unit scope).

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Holds and engineering queries

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Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

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1 · Equipment list

Tag	Name	Type	Size / rating	Side	Op. P kPa g / T °C	Design P kPa g / T °C	MDMT °C	Material	CA mm	Insulation	Thickness / rating	Weight t (empty / oper. / test)	Scope	Package
D-400	Flash drum	Vertical pressure vessel, 2:1 ellipsoidal heads, mesh-pad demister, skirt	1600 mm ID × 4800 mm T/T; 10.72 m³	Shell	103 / 74	450 / 150	0	SA-516-70 (shell, heads), SA-105 nozzles	3.0	— (PP at the inlet piping)	shell 8 mm, heads 10 mm (min. after forming 8.5 mm)	3.29 / 5.81 / 14.01	This unit	PKG-01
E-400	Feed preheater (hot oil)	Shell and tube, TEMA BEU (U-tube bundle, removable to the west)	305 mm shell ID × 4000 mm; 16.8 m² (72 tube holes); duty 420 kW	Shell (hot oil) Tubes (feed)	500 / 180 630 / 74	1,000 / 205 820 / 205	0 0	SA-516-70 shell / channel, SA-179 tubes, SA-266-2 tubesheet	3.0	H 50 mm	shell 6 mm, channel 6 mm	0.86 / 1.08 / 1.16	This unit	PKG-02
P-400	Feed supply pump (interface)	Centrifugal, upstream unit	rated 22 m³/h; shut-off 650 kPa g; motor 7.5 kW (interface, assumed)	Casing	630 / 10	820 / 65	0	by the upstream unit	—	—	—	—	Upstream unit (interface only)	—
FCV-400	Feed let-down valve	Globe (angle body for flashing service), equal-% trim, R = 30, fail closed	2" body (trim to suit), C_v 18.2 at 100 % (model)	Inlet / outlet	550 / 74	820 / 205	0	WCB body, hardened trim (flashing)	3.0	—	class 150	—	This unit	PKG-03
TCV-401	Hot-oil 3-way valve	3-way diverting, linear, fail to full bypass	2" (total 36 t/h)	Body	500 / 180	1,000 / 205	0	WCB	1.5	H	class 300	—	This unit	PKG-03
LCV-400	Bottoms level valve	Globe, equal-% trim, fail closed (flashing outlet)	2" body, C_v 48 at 100 % (model)	Body	134 / 49	450 / 150	0	WCB	3.0	—	class 150	—	This unit	PKG-03
PCV-420	Vapour pressure-control valve	Globe (or butterfly), equal-%, fail open	4" body, C_v 271 at 100 % (model)	Body	103 / 49	450 / 150	0	WCB	3.0	—	class 150	—	This unit	PKG-03
PCV-430	Nitrogen blanket regulator	Self-acting pressure regulator, set 30 kPa g	1", C_v 1.5 (assumed); wide-open capacity 194 kg/h N_2	Body	700 / 28	1,000 / 65	0	CS / SS trim	1.0	—	class 150	—	This unit	PKG-03
PSV-400	D-400 relief valve	Conventional spring-loaded, closed bonnet, to flare	set 450 kPa g; orifice per the Relief tab (P&ID: HOLD)	Inlet	103 / 49	450 / 150	0	WCB body, SS trim	3.0	—	class 150 × 150	—	This unit	PKG-05
XV-408	Feed SIS shut-off valve	Ball, full bore, spring-return actuator, fail closed, tight shut-off (class V)	2"	Body	630 / 10	820 / 65	0	WCB	3.0	AC	class 150	—	This unit	PKG-04
XV-409	Hot-oil SIS shut-off valve	Ball, fail closed, fire-safe	2"	Body	500 / 180	1,000 / 205	0	WCB	1.5	H	class 300	—	This unit	PKG-04
XV-418	Bottoms SIS shut-off valve	Ball, fail closed, tight shut-off (gas blow-by)	2" (3" with the line, query)	Body	134 / 49	450 / 150	0	WCB	3.0	—	class 150	—	This unit	PKG-04

Operating values are the highest of the three HMB cases (design, maximum rate 20 m³/h, turndown 10 m³/h); for everything that P-400 can pressurise the P-400 shut-off is the maximum operating pressure.

2 · Design-basis rules

• (1) Design pressure = max(1.10 × maximum operating pressure, maximum operating pressure + 170 kPa), rounded up to 10 kPa. The maximum operating pressure is the highest of the three HMB cases; for everything that P-400 can pressurise (feed line, E-400 tubes, let-down valve inlet) it is the P-400 shut-off 650 kPa g.

• (2) The D-400 system (drum, let-down outlet, vapour and bottoms lines to their first spec break) takes the MAWP on the P&ID, 450 kPa g = PSV-400 set; it must not be lower than rule (1) and the PSV set must not exceed it (ASME VIII UG-134).

• (3) Utilities: hot-oil system 1000 kPa g, nitrogen upstream of PCV-430 1000 kPa g, flare tail pipe 350 kPa g (assumed system design pressures).

• (4) The wall thickness adds the static head of liquid with D-400 liquid-full to the top tangent (overfill) at the bottom of the drum (UG-22).

• (a) Design temperature = maximum operating temperature + 25 °C, rounded up to 5 °C.

• (b) Everything that the hot oil can heat with the feed stopped (both sides of E-400, the hot feed line to FCV-400) takes the hot-oil design temperature 180 + 25 = 205 °C.

• (c) The D-400 system takes the P&ID design temperature 150 °C (it covers rule (a) on the 74 °C inlet and the 102 °C hot feed with TCV-401 wide open).

• (d) Hydrocarbon equipment and lines in the sun: not below 65 °C (solar gain, assumed).

• (e) Minimum design metal temperature (MDMT) 0 °C: the cold feed (10 °C) less a 10 K margin. Depressuring cannot chill the metal further: the feed boils at 20.1 °C and the drum liquid at 26 °C at atmospheric pressure (model VLE).

3 · Engineering queries

No.	Subject	Query	Why it matters	Proposed answer	Status
EM-01	D-400 elevation datum	The model's K.Hel = 3 m is the static height of the drum bottom (lowest point) above grade, the plot plan gives the bottom tangent at 3 m. The 2:1 head is 0.4 m deep, so one of them is 0.4 m off.	Static head to LCV-400 changes by ≈ 2.4 kPa; NLL elevation for the fire case changes by 0.4 m.	Fix the datum on the GA: bottom tangent 3 m above grade (plot plan), drum bottom 2.6 m; the model keeps 3 m.	OPEN
EM-02	"MAWP 450 kPa g" on the P&ID	The vessel calculated with the selected plates (shell 8 mm, heads 10 mm) has a MAWP of 684 kPa g (hot and corroded, top). The P&ID calls 450 kPa g the MAWP; it is the design pressure.	The stamped MAWP and the PSV set must be consistent (UG-134).	Stamp the vessel at the design pressure 450 kPa g @ 150 °C (common practice) and state the calculated MAWP on the datasheet; PSV-400 set 450 kPa g.	OPEN
EM-03	D-400 nozzles larger than the P&ID lines	Nozzle N1 (feed) is 6" and N3 (liquid) 3" in the Design tab sizing; the P&ID lines are 2". Nozzle N2 (vapour) 4" matches 4"-PV-4021-A1A.	A nozzle sized for the duty with a line two sizes smaller moves the restriction into the line.	Align with EP-01 / EP-03 (let-down outlet and bottoms line sizes) and show the reducers on the next P&ID revision.	OPEN
EM-04	E-400 thermal design by the vendor	The P&ID gives only the duty (417 kW); the model gives UA 3.89 kW/K clean. The datasheet assumes U 300 W/m² K (Design tab) and a BEU bundle with 16.8 m² (22 % excess at design).	Area, passes and baffles set the tube-side ΔP that the supply curve relies on (40 kPa).	Issue the datasheet for a vendor rating; ask for the clean / fouled U and the ΔP at the maximum rate.	OPEN
EM-05	No vent / purge connection on D-400	The P&ID shows a drain (HV-415) but no vent, steam-out or utility connection on D-400.	The drum must be gas-freed for entry and purged with N_2 before start-up.	Add a 2" vent with blind on the top head (V1) and a utility connection; route the vent to the flare.	OPEN